Ugo Bruzadin Nunes, PhD

CURRICULUM VITAE

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SUMMARY

Computational neuroscientist and machine-learning researcher with 8+ years of experience applying advanced statistics and machine learning to neural decoding, extracting insights from large datasets of neurophysiological time-series and behavioral data. Specialized in developing novel, faster methods to improve data processing and in using explainable AI to investigate how the brain works.

EDUCATION

Ph.D., Psychology (Cognitive Neuroscience) — Southern Illinois University, Carbondale — Aug 2023

Dissertation: "Short- and Long-Term Effects of Tobacco Abstinence, Bupropion, and Nicotine on Brain Activity During a Visuospatial Task."

Advisors: Dr. David Gilbert & Dr. Reza Habib

M.A., Psychology (Cognitive Neuroscience) — Southern Illinois University, Carbondale — Aug 2018

Advisors: Dr. Reza Habib & Dr. Matthew Schlesinger

B.A., Psychology — Pontifícia Universidade Católica de São Paulo, Brazil — Dec 2012

RESEARCH EXPERIENCE

Neuroscience Advisor — **NeuroLeap Corp.** Costa Mesa, CA (remote / part-time) — Jul 2026 – Present

- Advise executive leadership on brain–computer interface (BCI) paradigm design, data collection protocols, and hardware selection for neural decoding applications.
- Direct the technical development of real-time EEG signal processing pipelines, artifact rejection algorithms, and machine learning models for neural time-series data
- Evaluate emerging neuro-AI architectures and cross-modal machine learning frameworks to establish technical R&D milestones and product feasibility.
- Author technical documentation, experimental specifications, and research reports to guide engineering team execution and support investor/business development collateral.

Postdoctoral Fellow & Visiting Scholar — Brain Institute, Chapman University, Orange, CA — Sep 2023 – Present

Identified novel neural patterns predictive of conscious states across multiple tasks using advanced statistics, cluster analysis, machine learning, and neural networks.

- Managed a cross-university, multi-laboratory project (Chapman, Reed College, Tel Aviv), including clinical data collection (Unix, Git, SQL), personnel training, bi-monthly presentations, grant writing, and manuscript writing.
- Designed a 3D deep convolutional neural network (Keras, TensorFlow, Scikit-Learn) to classify conscious states from EEG signals, outperforming baseline models (XGBoost, SVM, Random Forest) by up to 12.48%.
- Developed a temporal generalization function that performs moving-window, stratified group cross-validation using parallel computing — 134% faster and more customizable than the MNE-Python baseline.
- Dataset: 3 EEG systems, 212 participants, 3 tasks, 4,000+ hours of EEG, 5+ TB of data.

Neural Data Scientist & ML Researcher — Alljoined Inc., San Francisco, CA — Apr 2025 – Jun 2026

- Built and equipped the company's in-house EEG research laboratory (32-channel rig with a real-time streaming stack); designed ML-focused experiments (image, video, and mental-imagery paradigms), wrote all data-collection protocols, and trained research assistants.
- Directed large-scale experimental loops on CLIP-based EEG-to-image frameworks — systematically testing layer freezing, alternative loss targets, and hyperparameter tuning to optimize latent-space alignment — and ran multi-stage ablation matrices to isolate performance variables.
- Improved cross-subject decoding accuracy up to 12% by engineering artifact-rejection, normalization, and feature-standardization pipelines robust to heterogeneous, non-stationary EEG; ran exploratory data analysis to surface distribution shifts, missingness, artifacts, and cohort biases across sessions and participants.

- Implemented supervised and self-supervised decoding models (PyTorch, Scikit-Learn) and conducted systematic failure analysis across noise regimes and edge cases; engineered parallelized bootstrapping and moving-window cross-validation to evaluate model stability and mitigate overfitting.
- Built and maintained the scalable experimental pipeline for a 1.6M+ trial multimodal dataset, from raw sensor ingestion to validated features; scaled training across GPU clusters (SLURM) with optimized batching and memory usage.
- Implemented LLM-driven semantic labeling workflows to scale creation of structured visual-stimulus datasets for cross-modal EEG decoding.
- Co-authored two works from this effort: the **Alljoined-1.6M** dataset — the largest consumer-grade EEG–vision dataset (arXiv preprint) — and **ENIGMA**, a state-of-the-art multi-subject EEG-to-image model presented as a poster at the NeurIPS 2025 Foundation Models for the Brain and Body Workshop.
- Partnered with engineering and product teams to translate exploratory findings and complex time-series analytics into production-ready demos used for fundraising and stakeholder presentations.
- Spearheading an image-to-emotion and emotion-to-preference generalization initiative (EEG-to-image and EEG-to-video) — owning experimental design, ML modeling, and product integration for an AI-driven neuromarketing product.

Machine Learning Neuroscience Consultant — Alljoined Inc., San Francisco, CA Feb – Mar 2025

- Designed and implemented an ML pipeline for EEG-based visual image reconstruction via EEG-to-CLIP feature alignment, reaching up to 70% accuracy on raw data (Git, PyTorch, Scikit-Learn, SLURM, AWS) in a multidisciplinary team.
- Led end-to-end EEG and mental-imagery experiment design and multi-device hardware synchronization (PsychoPy, C++, Lab Streaming Layer), aligning data capture with model-training requirements.
- Built automated feature-quality assessment and signal-to-noise metrics to neutralize real-world sensor noise and improve downstream model performance.

Graduate Researcher — Integrative Neuroscience Lab, Southern Illinois University, Carbondale, IL May 2018 – Jul 2023

Advisor: Dr. David Gilbert

EEG research on the short- and long-term neurocognitive effects of tobacco, bupropion, and nicotine during a spatial working-memory task (128-channel EEG; 116 participants across 12 longitudinal sessions; 8+ TB).

- Key findings: tobacco significantly slows brain frequency and power even after 66 days of abstinence, irrespective of treatment (sLORETA, cluster analysis); bupropion diverges from tobacco and the nicotine patch in parietal and prefrontal theta, alpha, and beta activity (GLM, rmMANOVA).
- Collected and managed longitudinal EEG and behavioral data; trained and supervised undergraduate research assistants who ran experimental sessions and performed first-pass analysis.
- Built and maintained the lab's EEG preprocessing, plotting, and analysis software — authoring QuickLab (~500% faster preprocessing; integrated CUDA-ICA for parallel computation) and the Inverse ICA plugin.
- Implemented a Random Forest model to decode gender, age, and smoking status from EEG signals (Python, Scikit-Learn).
- Developed an HTML/JavaScript experiment (5 tasks and questionnaires) with SQL-backed data capture into R, collecting from 100+ participants online and in person.

Senior Thesis Advisor — Psychological Laboratory, Webster University, Saint Louis, MO May 2022 – Jul 2023

- Advised a thesis on the effects of pink noise on sleep and anxiety; developed an app for remote participant exposure and data collection.
- Twice awarded the President's Student/Faculty Collaborative Research Grant.

Graduate Researcher — Cognition Lab, Southern Illinois University, Carbondale, IL Aug 2016 – Aug 2018

Advisor: Dr. Reza Habib

Behavioral research on mental imagery, mental rotation, and visual working memory.

- Designed and programmed browser-based behavioral experiments (JavaScript, HTML5, CSS); collected and analyzed behavioral data in R.
- Coordinated participant recruitment and data collection for the master's thesis; supported lab experiment design and analysis.

Graduate Researcher — Vision Lab, Southern Illinois University, Carbondale, IL Aug 2014 – Aug 2016

Advisor: Dr. Matthew Schlesinger

Investigated whether mental rotation and manual object-tracking share a common underlying mechanism.

- Collected and analyzed visual-tracking and behavioral data (E-Prime, SPSS).
- Trained and supported undergraduate students in data collection, analysis, and conference-poster preparation.

PUBLICATIONS

Peer-Reviewed & Preprint Papers

- Kneeland, R., Jiang, W., **Bruzadin Nunes, U.**, Scotti, P. S., Delorme, A., & Xu, J. (2026). ENIGMA: EEG-to-Image in 15 Minutes Using Less Than 1% of the Parameters. *Transactions on Machine Learning Research (TMLR)*, in press. arXiv:2602.10361.
- Xu, J., **Bruzadin Nunes, U.**, Jiang, W., Ryther, S., Pringle, J., Scotti, P. S., Delorme, A., & Kneeland, R. (2025). *Alljoined-1.6M: A Million-Trial EEG-Image Dataset for Evaluating Affordable Brain-Computer Interfaces*. arXiv preprint arXiv:2508.18571. Dataset: Alljoined-1.6M on Hugging Face.

Manuscripts Under Review & In Preparation

- **Bruzadin Nunes, U.**, Mudrik, L., Pitts, M., & Schurger, A. (2026). A Potential Novel Neural Correlate of Consciousness Shown in a No-Report Inattentional Blindness Paradigm. Manuscript in preparation.
- Wagganer, M., & **Bruzadin Nunes, U.** (2026). Effects of Pink Noise on Sleep and Anxiety. *Journal of Sleep Research*. Manuscript under review.
- **Bruzadin Nunes, U.**, Gilbert, D. G., Rabinovich, N., & Lindt, J. (2026). Tobacco Use Slows the Brain During a Working Memory Task Even After 66 Days of Abstinence. Manuscript in preparation.
- **Bruzadin Nunes, U.**, Gilbert, D. G., Rabinovich, N., & Lindt, J. (2026). The Differences Between Bupropion and Nicotine Replacement on Source-Localized Frequency Activation in Neural Networks. Manuscript in preparation.

Theses & Dissertation

- **Bruzadin Nunes, U.** (2023). Effects of Bupropion and Nicotine on Theta, Alpha, and Beta Regional Brain Source Activation During a Working Memory Task in Abstinent Smokers. Doctoral dissertation. opensiuc.lib.siu.edu/dissertations/2471.
- **Bruzadin Nunes, U.** (2018). Mental Imagery and Tracking. Master's thesis. opensiuc.lib.siu.edu/theses/2471.
- **Bruzadin Nunes, U.** (2012). A Queda do Patriarcado: A Crise da Masculinidade Contemporânea *(The Fall of the Patriarchy: A Crisis of Contemporary Masculinity)*. Undergraduate thesis. repositorio.pucsp.br.

SOFTWARE & OPEN-SOURCE PUBLICATIONS

- **Bruzadin Nunes, U.** (2024). Triangulation Project: a demonstration of the machine-learning method used to triangulate the neural correlates of consciousness. github.com/UgoBruzadin/Triangulation_Project.
- **Bruzadin Nunes, U.** (2024). QuickLab: an advanced program for EEG data visualization and preprocessing. EEGLAB plugin and github.com/UgoBruzadin/QuickLab.
- **Bruzadin Nunes, U.** (2024). Inverse ICA: an easy way to import and export ICA components from EEG files. EEGLAB plugin and github.com/UgoBruzadin/Inverse_ICA.

CONFERENCE POSTERS

- Kneeland, R., Jiang, W., **Bruzadin Nunes, U.**, Lee, S. K., Scotti, P. S., Delorme, A., & Xu, J. (2025). *ENIGMA: A Unified Lightweight EEG-to-Image Model for Multi-Subject Visual Decoding*. Poster, NeurIPS 2025 Foundation Models for the Brain and Body Workshop.
- **Bruzadin Nunes, U.**, Nicolacoudis, A., Sarig, A., Fish, N., Mudrik, L., Pitts, M., & Schurger, A. (Jun 2025). Two, Not One: Electrophysiological Correlates of Consciousness in a No-Report Inattentional Blindness Paradigm. Association for the Scientific Study of Consciousness (ASSC) Annual Meeting, Heraklion, Crete, Greece.
- Quach, S., Schurger, A., & **Bruzadin Nunes, U.** (Jan 2024). Triangulating Neural Correlates of Consciousness: Stability Analysis. Chapman University Student Scholar Symposium, Orange, CA.
- Wagganer, M., & **Bruzadin Nunes, U.** (Jan 2022). Sound of Sleep: The Effects of Pink Noise on Sleep Quality and Levels of Anxiety. Research Across Disciplines (RAD) Conference, Webster, MO.
- **Bruzadin Nunes, U.**, & Habib, R. (Apr 2019). Mental Imagery and Tracking. Midwestern Psychological Association, Chicago, IL.
- **Bruzadin Nunes, U.**, Russell, R., & Schlesinger, M. (Oct 2015). Does Mental Rotation and Manual Tracking Share an Underlying Mechanism? Society for Neuroscience, Chicago, IL.
- **Bruzadin Nunes, U.**, Schlesinger, M., & Russell, R. (Apr 2015). Visual Imagery, Mental Rotation, and Visual Working Memory: Tracing Paradigms. CURCA Graduate Forum, Southern Illinois University, Carbondale.
- **Bruzadin Nunes, U.** (Dec 2012). The Fall of the Patriarchate: The Crisis of Contemporary Masculinity. Undergraduate Final Dissertation Forum, Pontifícia Universidade Católica de São Paulo, Brazil.
- Batista, I. M., Nogueira, C. M., **Bruzadin Nunes, U.**, & Teixeira, T. (Apr 2011). Do Social Expectations Interfere in the Self-Image of Working Mothers? III Brazilian Congress of Psychology for Science and Profession, São Paulo, Brazil.

TALKS, PRESENTATIONS & GUEST LECTURES

- **Bruzadin Nunes, U.**, Nicolacoudis, A., Sarig, A., Fish, N., Mudrik, L., Pitts, M., & Schurger, A. (May 2025). Visual Awareness Positivity: A Novel Neural Correlate of Consciousness. Talk, Vision Sciences Society (VSS) Annual Meeting, St. Pete Beach, FL.
- Quach, S., **Bruzadin Nunes, U.**, & Schurger, A. (Jan 2024). Triangulating Neural Correlates of Consciousness: Stability Analysis. Presented by S. Quach, National Conference on Undergraduate Research (NCUR), Long Beach, CA.
- Wagganer, M., & **Bruzadin Nunes, U.** (Jan 2022). Sound of Sleep: The Effects of Pink Noise on Sleep Quality and Levels of Anxiety. Oral presentation by M. Wagganer, RAD Conference, Webster, MO.
- **Bruzadin Nunes, U.** (Jan 2024). Decoding Susceptibility to Inattentional Blindness. Research Seminar, Brain Institute.
- **Bruzadin Nunes, U.** (Aug 2023). Triangulating the Neural Correlates of Consciousness: A Pilot Study. Research Seminar, Brain Institute.
- **Bruzadin Nunes, U.** (Aug 2023). Source-Localizing the Neurocognitive Effects of Bupropion and NRT. Research Seminar, Brain Institute.
- **Bruzadin Nunes, U.** (Jan 2020 & Aug 2021). sLORETA-Assessed Frontal, Parietal, and Cingulate Effects of Bupropion vs. NRT vs. Placebo During a Spatial Working Memory Task (Dissertation Prospectus). Brain & Cognitive Sciences Pro-seminar.
- **Bruzadin Nunes, U.** (Jan 2019). Source Localization Algorithms for EEG/ERP. Brain & Cognitive Sciences Pro-seminar.
- **Bruzadin Nunes, U.** (Aug 2018). Mental Imagery and Tracking (Thesis Defense Update). Brain & Cognitive Sciences Pro-seminar.
- **Bruzadin Nunes, U.** (Jan 2018). Mental Imagery and Tracking (Thesis Prospectus Update). Brain & Cognitive Sciences Pro-seminar.
- **Bruzadin Nunes, U.** (Jan 2016). Mental Imagery and Tracking (Pilot Study Results). Brain & Cognitive Sciences Pro-seminar.
- **Bruzadin Nunes, U.** (Aug 2016). Psychology of Adolescence: A Brazilian Perspective. Guest lecture, Child Development.
- **Bruzadin Nunes, U.** (Aug 2015). Does Visual Imagery and Object Tracking Share an Underlying Mechanism? Brain & Cognitive Sciences Pro-seminar.
- **Bruzadin Nunes, U.** (Aug 2015). The Effects of THC on the Mind and Body. Guest lecture, Effects of Recreational Drugs.
- **Bruzadin Nunes, U.** (Jan 2015). How Can We Assess Mental Imagery? Brain & Cognitive Sciences Pro-seminar.
- **Bruzadin Nunes, U.** (Aug 2012). Psychology and Artificial Intelligence: Recent Contributions to Education. Panel discussion, Academic Week Symposium, Pontifícia Universidade Católica de São Paulo.
- **Bruzadin Nunes, U.** (Aug 2010). Jung's Studies on Memory and Dreams. Guest lecture, Analytical Psychology II.

HONORS, GRANTS & AWARDS

Triangulating Neural Correlates of Consciousness — Templeton World Charity Foundation Grant #30266 Sep 2022 – Aug 2025

- Amount awarded: $229,948. Director: Michael Pitts, Reed College. Role: Postdoctoral Fellow (data analysis, pre-registration editor, manuscript preparation). doi.org/10.54224/30266.

President's Student/Faculty Collaborative Research Grant — Faculty Sponsor (Awarded) Jan 2023

President's Student/Faculty Collaborative Research Grant — Faculty Sponsor (Awarded) Aug 2022

GPSC Graduate Teaching Award — Nominated Apr 2021

Menendez Award — Awarded for Research in Vision Jul 2019

GPSC Graduate Teaching Award — Nominated May 2019

Menendez Award — Awarded for Research in Vision May 2017

TEACHING EXPERIENCE

Visiting Assistant Professor — Psychology Department, Webster University Aug 2021 – May 2023

- Instructor of Record for 26 course sections, including:
  - Biopsychology — 2× (3-credit; Jan 2023, Jan 2022)
  - Careers in Psychology — 1× (1-credit; Jan 2023)
  - Careers in Psychology (asynchronous) — 4× (1-credit; 2× Jan 2022, Jan 2023)
  - Drugs and Chemical Dependency — 2× (3-credit; Jan 2023, Jan 2022)
  - Psychological Laboratory — 3× (1-credit; Jun 2022, Aug 2022, Jan 2023)
  - Psychology and Ethics (asynchronous) — 2× (2-credit; 2× Jan 2022)
  - Introduction to Measurements and Statistics — 4× (3-credit; Aug 2021, 2× Jan 2022, Aug 2022)

  - Introduction to Measurements and Statistics (asynchronous) — 1× (3-credit; Aug 2022)
  - Introduction to Psychology — 4× (3-credit; Aug 2021, Jan 2022, 2× Aug 2022)
  - Introduction to Psychology (asynchronous) — 2× (3-credit; Jun 2022, Aug 2022)
  - Introduction to Research Methods (asynchronous) — 1× (3-credit; Aug 2022)
- Supervised a student's senior thesis; taught in-person, online, and overload sections.
- Developed and updated course curricula with current research and materials.
- Collaborated with Dr. Andrew Elvington to provide Biopsychology students with a cadaver-based brain-dissection and hands-on neuroanatomy lab.

Instructor of Record — Southern Illinois University, Carbondale Aug 2017 – May 2021

- Instructor of Record for 9 three-credit courses, including:
  - Sensation and Perception — 4× (Jan 2021; Aug 2017 – May 2019)
  - Cognitive Psychology — 1× (Aug 2020)
  - Effects of Recreational Drugs — 4× (Aug 2019, Jan 2020, Jan 2016, Aug 2016)
- Developed and delivered in-person and online courses; managed TAs for grading and student communication.
- Designed syllabi, quizzes, exams, and research reports; adapted Cognitive Psychology to an online format (Zoom).
- Updated content with current research and interactive assignments (YouTube reports, in-person and online experiments).

Teaching Assistant — Southern Illinois University, Carbondale Jan 2015 – Jul 2020

- TA for 13 three-credit courses/sections, including:
  - Introduction to Psychology — 7× (Jun 2020, 2× Aug 2014, 2× Jan 2015, 2× Jun 2016)
  - Careers in Psychology — 3× (Aug 2015, 2× Jan 2017)
  - Psychology of Learning — 1× (Jun 2017)
  - Effects of Recreational Drugs — 2× (Jan & Aug 2015)
- Graded assignments, exams, and papers; assisted with syllabus development and classroom activities.
- Provided student support via office hours and email; proctored exams and supported in-person and online sessions.

English as a Second Language Teacher — Cia. Cultural, São Paulo, Brazil Jan – Jul 2014

- Taught English to a high-school class and a primary-school class; prepared lectures and created and graded activities, exams, quizzes, and papers.

Teaching Assistant, Analytical Psychology (Jung I & II) — Pontifícia Universidade Católica de São Paulo, Brazil Jan 2010 – Dec 2011

- Graded papers, took attendance, and proctored exams; guest lectured and held review sessions outside class.

MENTORSHIP & STUDENT ADVISING

- **Nick Fish**, BA (Aug 2024) — Triangulating Neural Correlates of Consciousness: P300 Analysis. Poster, Society for Neuroscience, Chicago, IL.
- **Sarah Quach**, BA (Aug 2023 – May 2024) — Triangulating Neural Correlates of Consciousness: Stability Analysis. Senior thesis (poster and oral presentation), Chapman University.
- **Maribeth Wagganer**, BA (Aug 2022 – May 2023) — Senior thesis, poster, oral presentation, and manuscript writing, Webster University.
- **Adam Holbrook**, MS (Jan 2021) — Nets 4 Nets. Class project presented at Southern Illinois University, Carbondale.
- **Eduardo Caminha**, MD (Jan 2015) — Examining the Effects of Multimodal Feedback on Skill Acquisition. Poster, Undergraduate Creative Activities and Research Forum, SIU Carbondale.
- **David Mersman**, MS (Jan 2015) — Does Mindfulness Predict the Ability to Shift Cognitive Focus? Poster, Undergraduate Creative Activities and Research Forum, SIU Carbondale.
- **Leticia Russell**, BS (Jan 2015) — Examining the Influence of Shifting Focus-of-Attention on Motor-Skill Performance. Poster, Undergraduate Creative Activities and Research Forum, SIU Carbondale.

ACADEMIC ADMINISTRATIVE EXPERIENCE & SERVICE

Administrative Graduate Assistant — School of Psychological & Behavioral Sciences, SIU Carbondale Jun 2017, 2018 & 2019

- Assessed and reorganized offices and laboratories for future use; archived key documents and securely destroyed confidential records; organized redistribution of academic hardware, journals, and books.

President — Cognitive Student Graduate Organization (CSGO), SIU Carbondale Jan – Dec 2016

Vice-President — Cognitive Student Graduate Organization (CSGO), SIU Carbondale Jan – Dec 2015

- Invited professors as guest lecturers for Brain & Cognitive Sciences students; helped create Brain Awareness Month; organized budgets, lodging, meetings, and events.

Departmental Assistant & Academic Tutor — Escola Mobile (High School), São Paulo, Brazil Jan 2012

- Observed and collected in-class data on student and teacher behavior (Excel, SPSS); participated in parent-teacher conferences; tutored students and helped with study organization.

CLINICAL EXPERIENCE

Phenomenological Psychotherapist — Clínica Ana Maria Popovich, São Paulo, Brazil Jan – May 2014

- Supervisor: Prof. Hélio Deliberador. Provided therapy for two clients during the semester.

Online Psychological Adviser — NPPI (Psychology and Internet Nucleus), São Paulo, Brazil Jan 2012 – May 2013

- Supervisor: Prof. Dr. Ivelise Fortim. Provided psychological orientation to the Brazilian community by email.

Psychotherapist — Hospital do Rim (Kidney Transplant Hospital), São Paulo, Brazil Jan – Dec 2012

- Supervisor: Prof. Dr. Maria Cecilia Roth. Offered in-bed cognitive behavioral therapy for clients in need.

Neurocognitive Therapist for Alzheimer's — ABraz (Brazilian Alzheimer Association), São Paulo, Brazil Aug – Dec 2011

- Supervisor: Prof. Fernanda Gouveia Paulino. Offered therapeutic exercises to eight patients focused on preserving cognition.

Career Counselor — Colégio Rainha da Paz (High School), São Paulo, Brazil Jan – May 2011

- Supervisor: Prof. Patricia Mortara. Delivered a career-advising course to 30 GED students.

Psychodiagnostician — Clínica Ana Maria Popovich, São Paulo, Brazil Jan – May 2011

- Supervisor: Prof. Luisa de Oliveira. Performed psychodiagnosis of a child client using TAT, WISC-III, and a parent interview.

RELEVANT GRADUATE COURSEWORK

- Aug 2014 – May 2023: Practicum — Teaching of Psychology.
- Aug 2014 – May 2023: Brain & Cognitive Sciences Pro-seminar.
- Jan 2020: Clinical Neuroanatomy.
- Jan 2019: Psychopharmacology & Behavior.
- Aug 2018: Readings in Psychology — R for Data Analysis.
- Aug 2017: Advanced Topics in Developmental Psychology.
- Jan 2017: Readings in Psychology — R for Data Presentation.
- Aug 2016: Behavioral Genetics; American Idealism (Philosophy).
- Jan 2016: Human Learning & Memory; Life-Span Developmental Psychology.
- Aug 2015: Behavioral Ecology (Seminar); Multivariate Methods of Psychology.
- Jan 2015: Aging, Memory & Cognition; Experimental Design and Analysis.
- Aug 2014: History & Systems of Psychology; Neurobiological Bases of Behavior.
- Neurobiology of Learning and Memory (Seminar).

PROFESSIONAL TRAINING & CERTIFICATIONS

- **Machine Learning** — Coursera (Stanford University), Instructor: Andrew Ng. Completed 2024.
- **Deep Learning Specialization** — Coursera (DeepLearning.AI). Completed 2024.
- **Neural Networks and Data Processing with Python** — Kaggle Learn. Completed 2024.
- **Linear Algebra** — MIT OpenCourseWare, Instructor: Prof. Gilbert Strang. Completed 2023.
- **Workshop: "The Productive Use of Space in Sign Language"** by Dr. Scott Liddell — Universidade de São Paulo, Brazil. Oct 2013.
- **Workshop: "Retinal Strategies of Visual Coding"** by Dr. Cristina Joselevich — Instituto de Psicologia, Universidade de São Paulo, Brazil. Aug 2012.

LANGUAGES

- Portuguese — Native / Fluent.

- English — Fluent (listening, reading, speaking, writing).
- Spanish — Intermediate (listening, speaking, reading); Novice (writing).
- French — Intermediate (listening, reading); Novice (speaking, writing).

TECHNICAL SKILLS

- **Programming & Tools:** Python (PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Pydantic), MATLAB, R, SQL, C/C++, JavaScript, HTML5/CSS, Unix, SPSS, E-Prime.
- **Neural Data & BCI:** EEG/MEG decoding, Lab Streaming Layer (LSL), artifact rejection, feature extraction, time–frequency analysis, signal-to-noise optimization, adaptive & cross-modal decoding.
- **Machine Learning:** deep CNNs, transformers, RNNs, self-supervised learning, latent-space alignment (CLIP), attention & reinforcement models, tree-based models, time-series forecasting, interpretability (SHAP).
- **Statistical Modeling:** mixed-effects models, bootstrap methods, Bayesian inference, generalized linear models (GLM), rmMANOVA, multivariate analysis.
- **Infrastructure & Reproducibility:** SLURM cluster orchestration, AWS, Docker, Git, experiment versioning.
- **Data Visualization:** Matplotlib, Seaborn.

OTHER

U.S. and Brazilian citizen.

Academic interests: brain-computer interfaces, computational neuroscience, philosophy of mind, metaphysics, linguistics, 3D printing, and game design.